

IMO 2026 — Problem 6

Solution by Claude Fable 5

2026-07-16

Source: `results/imo-2026/claude-fable-5/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved

Approaches tried

- **small-prime-core** — SOLVED (proof-reviewer APPROVE, round 3). Fix prime universe $Q = \{p \mid a_1 \mid p\}$; SCPL by minimal-counterexample descent (companion lemma, threshold $z = a_1$); collapse constraints mod $M = Q$; greedy enumeration of exactly-periodic set. Complete, no gaps.
- **wqo-domination** — SOLVED (proof-reviewer APPROVE, round 3). Independent SCPL descent; minimal constraint family E_∞ finite 2^Q ; V_∞ exactly L -periodic for $L = (E_\infty)$; greedy enumeration of periodic set from $n=1$. Complete, no gaps.
- **greedy-clique-closure** — earlier scaffold (round 2), superseded; SCPL crux left open there.

Both solved approaches were independently verified by the reviewer: the SCPL descent (crux) is well-founded (strong induction on the larger index j ; minimal-counterexample on the smaller index i ; companion x lands as an earlier term by valid-below-are-terms + greedy minimality, forcing a small shared prime). SCPL, the companion construction, and periodicity-from- $n=1$ were confirmed computationally for $a_1 \in \{2, 3, 15, 26, 30, 32, 48, 49, 77, 105, 210\}$ (including the non-squarefree case $a_1=48$, term $56=2^3 \cdot 7$, where the threshold $z=a_1$ is load-bearing).

Current best

Complete rigorous proof of the full claim (below), with explicit T,L.

Full proof

Problem. Let $a_1, a_2, a_3, \dots$ be an infinite sequence of integers greater than 1 such that for every positive integer n , a_{n+1} is the smallest integer greater than a_n satisfying $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, \dots, n$. Prove that there exist positive integers T and L with $a_{n+T} = a_n + L$ for every positive integer n .

Notation

- $P(m)$ — the set of prime factors of an integer $m > 1$.
- $Q = \{p \text{ prime} : p \leq a_1\}$ — the **small primes**. Since $a_1 > 1$, $2 \in Q$, so $Q \neq \emptyset$; Q is finite.
- $\sigma(m) = P(m) \cap Q = \{p \text{ prime} : p \mid m, p \leq a_1\}$ — the small prime factors of m .
- $M = \prod_{p \in Q} p$ (≥ 2).
- $B = \text{rad}(a_1) = \prod_{p \in P(a_1)} p$.
- $W_n = \{m \in \mathbb{Z} : m > 1, \gcd(m, a_i) > 1 \text{ for all } 1 \leq i \leq n\}$ (no lower bound beyond $m > 1$).
- $V_\infty = \{m \in \mathbb{Z} : m > 1, \gcd(m, a_i) > 1 \text{ for all } i \geq 1\}$.
- $A = \{a_1, a_2, \dots\}$.

The defining rule restates as: for every $n \geq 1$,

$$a_{n+1} = \min(W_n \cap (a_n, \infty)). \quad (*)$$

(For $m > a_n \geq a_1 > 1$ the condition $m > 1$ is automatic, so $(*)$ is literally the rule.) The sequence is strictly increasing, so $a_n \geq a_1 + (n-1)$ and $a_n \rightarrow \infty$; every integer $x \geq a_1$ is either a term or lies in a unique gap $a_n < x < a_{n+1}$.

Lemma 1 (Clique). For all $m \neq n$, $\gcd(a_m, a_n) > 1$.

Proof. WLOG $m > n$. By $(*)$, $a_m \in W_{m-1}$ and $n \leq m-1$, so $\gcd(a_m, a_n) > 1$. $\square$

Lemma 2 (Bounded gaps / well-posedness). For every n , every multiple of B exceeding 1 lies in W_n ; hence $W_n \cap (a_n, \infty) \neq \emptyset$ and $a_{n+1} \leq a_n + B$.

Proof. Let $m > 1$, $B \mid m$, $i \leq n$. By Lemma 1 (trivial if $i = 1$), $\gcd(a_i, a_1) > 1$, so a prime p divides a_i and a_1 ; then $p \mid B \mid m$, so $\gcd(m, a_i) > 1$. Thus $m \in W_n$. The least multiple of B above a_n is $\leq a_n + B$ and exceeds $a_n \geq a_1 > 1$, so lies in $W_n \cap (a_n, \infty)$; by $(*)$, $a_{n+1} \leq a_n + B$. $\square$

Lemma 3 (Reduction). (i) $A \subseteq V_\infty$. (ii) $a_{n+1} = \min(V_\infty \cap (a_n, \infty))$. (iii) $A = V_\infty \cap [a_1, \infty)$, and the sequence is its increasing enumeration.

Proof. (i) $a_k > 1$; for $i = k$, $\gcd(a_k, a_k) = a_k > 1$; for $i \neq k$, Lemma 1. (ii) $V_\infty \subseteq W_n$, so $\min(V_\infty \cap (a_n, \infty)) \geq \min(W_n \cap (a_n, \infty)) = a_{n+1}$; and $a_{n+1} \in V_\infty \cap (a_n, \infty)$ by (i) gives $\leq a_{n+1}$. (iii) ($\subseteq$) by (i) and $a_n \geq a_1$. ($\supseteq$) If $v \in V_\infty$, $v \geq a_1$, $v \notin A$, then $a_n < v < a_{n+1}$ for some n (increasing, unbounded), so $v \in V_\infty \cap (a_n, \infty)$ with $v < a_{n+1}$, contradicting (ii). $\square$

Lemma 4 (Valid-below-are-terms). For every n : $W_n \cap [a_1, a_n] = \{a_1, \dots, a_n\}$.

Proof. Induction on n . Base $n = 1$: $[a_1, a_1] = \{a_1\}$, $a_1 \in W_1$. Step: assume for n . $(\supseteq)$ each a_k , $k \leq n + 1$, is in $[a_1, a_{n+1}]$ and W_{n+1} (Lemma 1). $(\subseteq)$ $x \in W_{n+1} \cap [a_1, a_{n+1}] \subseteq W_n$: if $x = a_{n+1}$ done; if $a_n < x < a_{n+1}$ then $x \in W_n \cap (a_n, \infty)$, $x < a_{n+1}$, contradicting $(*)$; if $a_1 \leq x \leq a_n$ then $x \in W_n \cap [a_1, a_n] = \{a_1, \dots, a_n\}$ by IH. $\square$

Lemma 5 (Companion). Let $a \in A$ have a prime factor $q > a_1$. Then $\sigma(a) \neq \emptyset$ and there is an integer x with $P(x) = \sigma(a)$ and $a_1 \leq x < a$.

Proof. $\sigma(a) \neq \emptyset$: $\gcd(a, a_1) > 1$ (Lemma 1, trivial if $a = a_1$) gives a prime $p_0 \mid a, a_1$, and $p_0 \leq a_1$, so $p_0 \in \sigma(a)$. Fix $p \in \sigma(a)$; $p \leq a_1 < q$, so $p \neq q$, $pq \mid a$, whence $a \geq pq$ (5.1). Let $m_0 = \prod_{r \in \sigma(a)} r$ (squarefree, $P(m_0) = \sigma(a)$). Choose $t \geq 0$ minimal with $m_0 p^t \geq a_1$, set $x = m_0 p^t$; then $P(x) = \sigma(a)$ and $x \geq a_1$. If $t = 0$: $x = m_0 \mid \text{rad}(a) \mid a$, and $q \nmid m_0$ (as $q > a_1$) so $m_0 q \mid a$, giving $a \geq m_0 q > m_0 = x$. If $t \geq 1$: minimality gives $m_0 p^{t-1} < a_1$, so $x = p(m_0 p^{t-1}) < p a_1 < pq \leq a$ by (5.1). $\square$

(Threshold is load-bearing: with $z = \text{rad}(a_1)$ or $z = \max P(a_1)$ the lemma fails for $a_1 = 48$, term $56 = 2^3 \cdot 7$; with $z = a_1$ the prime 7 is small and no companion is needed. Only finiteness of Q is used downstream.)

Lemma 6 (Small Common Prime Lemma, SCPL). For all $i < j$: $\sigma(a_i) \cap \sigma(a_j) \neq \emptyset$.

Proof. Strong induction on j ; $S(j)$: “ $\sigma(a_i) \cap \sigma(a_j) \neq \emptyset$ for all $i < j$.” $S(1)$ vacuous. Fix $j \geq 2$, assume $S(j')$ for all $j' < j$. Suppose some $i < j$ has $\sigma(a_i) \cap \sigma(a_j) = \emptyset$; take i **minimal**. - (6.1) For $s < i$: $\sigma(a_s) \cap \sigma(a_j) \neq \emptyset$ (minimality of i). - (6.2) $\gcd(a_i, a_j) > 1$ (Lemma 1) gives a common prime q ; $q \leq a_1$ would contradict the choice of i , so $q > a_1$. - $i \geq 2$: if $i = 1$, a common prime r of a_1, a_j lies in $P(a_1) = \sigma(a_1)$ and in $\sigma(a_j)$, contradiction.

Step 1. By (6.2), Lemma 5 applied to a_i gives x with $P(x) = \sigma(a_i)$, $a_1 \leq x < a_i$ (6.3). **Step 2** ($x \in W_{i-1}$). Since $i < j$, $S(i)$ holds by the IH: for $i' \leq i - 1$ pick $r \in \sigma(a_{i'}) \cap \sigma(a_i)$; then $r \mid a_{i'}$ and $r \mid x$ (as $r \in \sigma(a_i) = P(x)$), so $\gcd(x, a_{i'}) > 1$. Also $x > 1$. **Step 3** (x is an earlier term). $i - 1 \geq 1$. If $a_{i-1} < x < a_i$: $x \in W_{i-1} \cap (a_{i-1}, \infty)$, $x < a_i$, contradicting $(*)$. Hence $a_1 \leq x \leq a_{i-1}$, so by Lemma 4, $x = a_s$, $s \leq i - 1$. **Step 4 (contradiction).** By (6.1) with $s < i$: pick $r' \in \sigma(a_s) \cap \sigma(a_j)$. Then $r' \mid a_s = x$ so $r' \in P(x) = \sigma(a_i)$ (so $r' \leq a_1$, $r' \mid a_i$); and $r' \mid a_j$. Hence $r' \in \sigma(a_i) \cap \sigma(a_j)$, contradiction. Thus $S(j)$ holds. $\square$

Lemma 7. For every $a \in A$, $\sigma(a) \neq \emptyset$.

Proof. $a = a_1$: $P(a_1) = \sigma(a_1) \neq \emptyset$. $a = a_j$, $j \geq 2$: Lemma 6 on (a_1, a_j) . $\square$

Step 5: The term set is determined by residues mod M

Let $\mathcal{F} = \{\sigma(a) : a \in A\}$, a finite family of nonempty subsets of Q (Lemma 7; $\mathcal{F} \subseteq 2^Q$).

Lemma 8. $A = \{m \geq a_1 : \sigma(m) \cap S \neq \emptyset \text{ for every } S \in \mathcal{F}\}$. *Proof.* $(\subseteq)$ $m = a_k \geq a_1$; for $S = \sigma(a_\ell)$: if $\ell \neq k$, Lemma 6; if $\ell = k$, Lemma 7. $(\supseteq)$ $m \geq a_1$ with the property: for each i , a prime $p \in \sigma(m) \cap \sigma(a_i)$ gives $\gcd(m, a_i) > 1$, so $m \in V_\infty \cap [a_1, \infty) = A$ by Lemma 3(iii). $\square$

Lemma 9. $m \equiv m' \pmod{M} \Rightarrow \sigma(m) = \sigma(m')$ (each $p \in Q$ divides M). Defining $\sigma(r)$ for

$r \in \mathbb{Z}/M\mathbb{Z}$ and $R = \{r : \sigma(r) \cap S \neq \emptyset \ \forall S \in \mathcal{F}\},$

$$A = \{m \geq a_1 : (m \bmod M) \in R\}. \quad (9.1)$$

R is a well-defined subset of the finite set $\mathbb{Z}/M\mathbb{Z}$, and $R \neq \emptyset$ since $a_1 \in A \Rightarrow (a_1 \bmod M) \in R$.

Step 6: Conclusion

Set $T = |R|$ (≥ 1), $L = M = \prod_{p \leq a_1} p$ (≥ 2). Fix $n \geq 1$. The window $(a_n, a_n + M]$ has M consecutive integers, hence exactly one per residue class mod M , all $\geq a_1$; by (9.1), $|A \cap (a_n, a_n + M]| = |R| = T$. Also $a_n + M \equiv a_n \pmod{M}$ with $a_n \in A$ and $a_n + M > a_1$, so $a_n + M \in A$. Since (a_k) enumerates A increasingly, $A \cap (a_n, a_n + M] = \{a_{n+1}, \dots, a_{n+T}\}$ with a_{n+T} its largest element; and $a_n + M$ is the largest integer in the window, hence the largest element of A there. Therefore

$$a_{n+T} = a_n + M = a_n + L \quad \text{for every } n \geq 1,$$

with $T = |R| \geq 1$ and $L = \prod_{p \leq a_1, \ p \text{ prime}} p \geq 2$. ■

(The identity holds from $n = 1$: no transient, since the enumeration of the exactly-periodic set A begins at $a_1 \in A$. **answer_type** is none; no numeric answer to verify. An independent second proof — **wqo-domination** — reaches the same conclusion via the inclusion-minimal constraint family $E_\infty \subseteq 2^Q$ and $L = \prod(\bigcup E_\infty)$.)